Scott T. Wieman, Ph.D.

MS #61 · David Center for Ocean Innovation 327
Applied Ocean Physics and Engineering Department
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Statistics

Publications:
First-author **6**
Total peer-reviewed **11**
h-index **5**
Citations **63**

Presentations:
Invited **3**
Contributed first-author **16**
Total contributed **37**

Fellowships and Grants:
Non-travel grants **5**
Fellowships **6**
Total amount **$206,900**

Teaching:
Courses developed **1**
Courses TAed **7**
Guest lectures **10**

Education

Massachusetts Institute of Technology, Cambridge, MA 2021-2026
and **Woods Hole Oceanographic Institution**, Woods Hole, MA
Doctor of Philosophy in Geochemistry, May 2026
Dissertation: *Methane Cold Seeps: Paleo Proxies, Modern Measurements, and Instrument Development*
Advisors: Weifu Guo and Anna Michel
Certificates: Graduate Teaching Certificate, June 2024; Research Mentoring Certificate, June 2025

University of Maryland Francis King Carey School of Law, Baltimore, MD 2018-2020
Master of Science in Law in Environmental Law, May 2020
Thesis: *A NEPA for the Modern World: Lessons for Adapting NEPA to Better Account for Greenhouse Gas Emissions*
Advisor: William Piermattei

Harvard University, Cambridge, MA 2015-2017
Master of Arts in Earth and Planetary Science, May 2017
Advisor: Daniel Schrag

Williams College, Williamstown, MA 2010-2014
Bachelor of Arts in Chemistry, June 2014

Research Appointments

Woods Hole Oceanographic Institution

Postdoctoral Investigator, *Applied Ocean Physics and Engineering* 2026-*present*
- Developing an in situ CO_2 sensor to spectroscopically measure $\delta^{13}C$ of DIC in deep ocean environments in order to investigate sources of CH_4 at marine cold seeps and its oxidation and transportation.
- Analyzing dissolved CH_4 and CO_2 in estuarine and coastal environments in New York and the Canadian Arctic to understand physical and biological controls on greenhouse gas emissions from these locations.

Graduate Student Researcher, *Marine Geology and Geophysics* 2021-2026
- Studied CH_4 dynamics at cold seeps through numerical modeling of porewater isotopes, clumped isotope measurements of seep carbonates, and field campaigns utilizing ROV-mounted CH_4 sensors for seep detection and sample collection (**Publications 10** & **11**).
- Developed prototype laser spectrometers for making optical clumped isotope measurements of CO_2 in both traditional gas cells and hollow core fibers (**Publications 5** & **8**).

National Aeronautics and Space Administration

Curiosity Rover Science Team Member 2018-2021
- Isotope geochemist for the Sample Analysis at Mars (SAM) instrument suite aboard rover *Curiosity* at Gale Crater, Mars.
- Studied $\delta^{15}N$ in Martian surface samples to elucidate past nitrogen cycling processes and develop a correction function for nitrogen isotopes on the SAM testbed at NASA's Goddard Space Flight Center (**Publication 9**).
- Studied $\delta^{13}C$, $\delta^{18}O$, and Δ_{47} in Martian surface samples, the Martian atmosphere, and terrestrial analogs.

Stable Isotope Facilities Manager, *Goddard Space Flight Center* 2017-2021
- Managed the ThermoFinnegan Delta V isotope ratio mass spectrometer and front ends in the Planetary Environments Laboratory.
- Measured bulk $\delta^{13}C$, $\delta^{15}N$, and δD in laboratory analogs of Martian clays and field analogs (**Publication 3**).
- Measured compound-specific $\delta^{13}C$ to understand carbon isotope fractionation associated with aerosol formation on Titan (**Publication 6**).
- Used an underwater membrane inlet mass spectrometer to measure $\delta^{13}C$ and $\delta^{18}O$ of CO_2 in the Gulf of Mexico.

Assistant Research Faculty, *Center for Space Sciences and Technology, University of Maryland, Baltimore County* 2017-2021
- Studied planetary isotope systems in a non tenure-track position hosted at NASA's GSFC.

Harvard University

Graduate Research Assistant, *Department of Earth and Planetary Sciences* 2015-2017

- Made clumped isotope measurements of carbonate accretionary lapilli from the Chicxulub impact cloud to examine upper temperature limits of the paleothermometer and the formation environment of the particulates.

Williams College

Research Associate, *Geosciences Department and Center for Environmental Studies* 2014-2015

- Designed and led a research project on aqueous chemical fluxes in Birch Brook and Ford Glen in Williamstown, MA and their implications for understanding acid rain buffering mechanisms of foliage, soil, and bedrock (**Publication 1**).

Undergraduate Research Assistant, *Chemistry Department and Center for Environmental Studies* 2013

- Developed an interdisciplinary project developing rapid extraction methods for total soil-bound phosphorus, a major factor in plant growth.

Undergraduate Research Assistant, *Chemistry Department* 2012

- Used cavity ringdown spectroscopy to study the fifth and sixth C-F stretching overtones in deuterated hydrofluorocarbons, important atmospheric pollutants.
- Performed density functional theory computations to model conformational geometries and related energy states.

Publications (63 citations, 5 h-index)

Invited:

Stern, J.C., **S.T. Wieman**. Traditional Stable Isotope Geochemistry in *Encyclopedia of Geology* (Academic Press, Cambridge, MA, ed. 2, vol. 5, pp. 100-113, 2021). https://doi.org/10.1016/B978-0-08-102908-4.00116-8

Peer-reviewed: (* denotes equal authorship)

11 **Wieman, S.T.**, W. Guo. Global high rates of anaerobic methane oxidation at marine cold seeps recorded by isotope bonding in seep carbonates. *Proceedings of the National Academy of Sciences of the USA* **123**, e2533639123 (2026). https://doi.org/10.1073/pnas.2533639123

10 **Wieman, S.T.**, et al. Quantifying deep ocean methane along the central Chilean Margin. *Geophysical Research Letters* **53**, e2026GL124079 (2026). https://doi.org/10.1029/2026GL124079

9 Andrejkovičová, S., **et al**. NH_4-smectites as potential precursors of nitrogen compounds evolved during analyses by the Sample Analysis at Mars (SAM) instrument on the Curiosity rover. *Journal of Geophysical Research Planets* **131**, e2025JE009610 (2026). https://doi.org/10.1029/2025JE009610

8 **Wieman, S.T.**, J. Kapit, W. Guo, A.P.M. Michel. Clumped isotope measurements on nanomoles of CO_2 using a hollow core fiber-based spectrometer. *Optics Express* **33**, 51820-51829 (2025). https://doi.org/10.1364/OE.582189

7 **Wieman, S.T.** First record of melanism in *Myiarchus antillarum* (Puerto Rican Flycatcher). *Journal of Caribbean Ornithology*, **38**, 34-38 (2025). https://doi.org/10.55431/jco.2025.38.34-38

6 Ugelow, M.S., **S.T. Wieman**, M.C.R. Schwarz, V. Da Poian, J.C. Stern, M.G. Trainer. Laboratory studies on the influence of hydrogen on Titan-like photochemistry. *ACS Earth and Space Chemistry*, **8**, 2362-2371 (2024). https://doi.org/10.1021/acsearthspacechem.4c00102

5 **Wieman, S.T.**, J. Kapit, A.P.M. Michel, W. Guo. Development of a Quantum cascade laser absorption spectrometer for simultaneous measurement of ^{13}C-^{18}O and ^{18}O-^{18}O clumping in CO_2. *Rapid Communications in Mass Spectrometry* **38**, e9836 (2024). https://doi.org/10.1002/rcm.9836

4 Mattison, K.A.*, A.R. Merchak*, **S.T. Wieman***, S. Zimmer*, S.C. Fankhauser. Engaging young scholars in science through publication: a survey analysis of published middle and high school authors. *Learned Publishing* **35**, 321-331 (2022). https://doi.org/10.1002/leap.1480

3 Murphy, A.E., **S.T. Wieman**, J. Gross, J.C. Stern, A. Steele, M. Glamoclija. Preservation of organic carbon in dolomitized Cambrian stromatolites and implications for microbial biosignature preservation in diagenetically replaced carbonate rock. *Sedimentary Geology* **410**, 105777 (2020). https://doi.org/10.1016/j.sedgeo.2020.105777

2 Neveu, M., C.H. House, **S.T. Wieman**. Phoebe's carbon isotope composition as evidence for self-shielding of ^{12}CO in the solar nebula. *Icarus* **345**, 113714 (2020). https://doi.org/10.1016/j.icarus.2020.113714

1 Dethier, D.P., **S.T. Wieman**, J. Racela. Thirty-year trends in acid deposition and neutralization in two headwater streams, northwestern Massachusetts. *Hydrological Processes* **32**, 3464 (2018). https://doi.org/10.1002/hyp.13270

White Papers and Reports:

FUTURE 2024 PI team, **et al**. The FUTURE of the US marine seafloor and subseafloor sampling capabilities. *AGU Advances* **6** (2025). https://doi.org/10.1029/2024AV001560

Panieri, G., **et al**. Leg 2: Cold Seeps Extreme24 Expedition Report. *Septentrio Reports* **1** (2025). https://doi.org/10.7557/7.7986

Kao, D.-Y., **et al**. Impacts of quantum chemistry calculations on exoplanetary science, planetary astronomy, and astrophysics. *Bulletin of the American Astronomical Society* **51**, 240 (2019). https://doi.org/10.48550/arXiv.1904.07161

Grants and Fellowships

COBRA Early Career Accelerator Award, Bigelow Laboratory for Ocean Sciences 2025
Teaching Development Fellowship, Massachusetts Institute of Technology Teaching and Learning Lab 2024
Ocean Ventures Fund Grant, Woods Hole Oceanographic Institution (Principal Investigator) 2024
Ocean Engineering and Instrumentation Ph.D. Fellowship, The Link Foundation 2023
Ocean Ventures Fund Grant, Woods Hole Oceanographic Institution (Principal Investigator) 2022
Fundamental Laboratory Research Grant, National Aeronautics and Space Administration (Co-Investigator) 2018
Make Our Planet Great Again Doctoral Fellowship, French Ministry of Europe and Foreign Affairs, *declined* 2018
Skaff Family Environmental Graduate Fellowship, Harvard University Center for the Environment 2015
Fulbright U.S. Student Research Fellowship, J. William Fulbright Foreign Scholarship Board, *declined* 2015
Graduate Research Grant, American-Scandinavian Foundation, *declined* 2015
Robert Ikemori Quay '04 Outing Club Memorial Fellowship, Williams College Outing Club 2014

Conference Travel Grants:
Office of Graduate Education Conference Travel Grant, Massachusetts Institute of Technology 2025
Graduate Student Travel Grant, Hebrew University of Jerusalem 2022
Graduate Student Council Conference Travel Grant, Massachusetts Institute of Technology 2022

Honors and Awards

Dean's Award for Excellence in the Masters of Science in Law, University of Maryland Francis King Carey School of Law 2020
Phi Kappa Phi, University of Maryland, Baltimore Campuses Chapter of Phi Kappa Phi 2020
Certificate of Distinction in Teaching (PHYSCI 11), Harvard University Derek Bok Center for Teaching and Learning 2017
Certificate of Distinction in Teaching (SPU 25), Harvard University Derek Bok Center for Teaching and Learning 2017
Dean's List, Williams College 2010-2011, 2013-2014
Class of 1960's Scholar in Chemistry, Williams College Chemistry Department 2013

First Author Presentations (second+ author: 21)

Invited:

Wieman, S.T., J. Beeson, A.D. Skarke, S. Youngs, J. Kapit, J.J. Marlow, A.P.M. Michel. Integrating Real Time Multibeam and In Situ Methane Measurements for Seep Hunting: A Case Study from the Chilean Margin. *Crustal Ocean Biosphere Research Accelerator Webinar Series*, virtual, oral presentation, Jan. 2026.

Wieman, S.T., J. Kapit, A.P.M. Michel, W. Guo. Towards the development of a rapid and precise isotope ratio laser spectrometer for dual clumped measurements of carbonates. *Aerodyne Research Seminar Series*, Billerica, Massachusetts, oral presentation, Sep. 2023.

Wieman, S.T. Light stable isotope research in support of ongoing and future NASA flight missions. *Geological Society of Washington*, Washington, District of Columbia, oral presentation, Sep. 2019.

Contributed:

Wieman, S.T., S. Youngs, S. Marquardt, J. Remar, J. Kapit, J.J. Marlow, A.P.M. Michel, Elevated methane levels in the Biobío Canyon of central Chile reveal offshore transport of organic carbon. *Ocean Sciences Meeting*, Glasgow, Scotland, oral presentation, Feb. 2026.

Wieman, S.T., W. Guo. Global rates of Anaerobic Methane Oxidation at Marine Cold Seeps Constrained by Carbonate Clumped Isotopes. *American Geophysical Union Fall Meeting*, New Orleans, Louisiana, oral presentation, Dec. 2025.

Wieman, S.T., J. Beeson, A.D. Skarke, S. Youngs, J. Kapit, J.J. Marlow, A.P.M. Michel. Integrating Real Time Multibeam and In Situ Methane Measurements for Seep Hunting: A Case Study from the Chilean Margin. *American Geophysical Union Fall Meeting*, New Orleans, Louisiana, poster presentation, Dec. 2025.

Wieman, S.T., J. Kapit, W. Guo, A.P.M. Michel. First detection of clumped isotopologues of CO_2 using a hollow-core waveguide. *Field Laser Applications in Industry and Research Conference*, Assisi, Italy, oral presentation, Sep. 2024.

Wieman, S.T., J. Kapit, A.P.M. Michel, W. Guo. A quantum cascade laser absorption spectrometer for simultaneous measurements of ^{13}C-^{18}O and ^{18}O-^{18}O clumping in CO_2. *Ninth International Clumped Isotope Workshop*, Stony Brook, New York, oral presentation, Aug. 2024.

Wieman, S.T., J. Kapit, A.P.M. Michel, W. Guo. Development of a Quantum-Cascade Laser absorption spectrometer for rapid and precise measurement of ^{13}C-^{18}O and ^{18}O-^{18}O Clumping in CO_2. *Conference on Lasers and Electro-Optics*, San Jose, California, oral presentation, May 2023.

Wieman, S.T., W. Guo. Controls on clumped isotope composition of cold seep carbonates: A potential quantitative proxy for the rate of anaerobic methane oxidation. *Eighth International Clumped Isotope Workshop*, Jerusalem, Israel, oral presentation, Sep. 2022.

Wieman, S.T., W. Guo. Controls on clumped isotope composition of cold seep carbonates: Implications for reconstructing the rate of anaerobic oxidation of methane. *American Geophysical Union Fall Meeting*, New Orleans, Louisiana, oral presentation, Dec. 2021.

Wieman, S.T., V. Da Poian, M.C. Roach, M.S. Ugelow, J.C. Stern, M.G. Trainer. The role of Hydrogen in isotopic fractionation and atmospheric composition during photochemical aerosol formation in a Titan-like atmosphere. *American Geophysical Union Fall Meeting*, New Orleans, Louisiana, oral presentation, Dec. 2021.

Wieman, S.T.. Accounting for indirect greenhouse gas emissions in the National Environmental Policy Act: A case study from the energy sector. *Graduate Climate Conference*, Virtual, poster presentation, Oct. 2021.

Wieman, S.T., M.S. Ugelow, V. Da Poian, M.C. Roach, J.C. Stern, M.G. Trainer. Effect of Titan-like aerosol formation on δ^{13}C of major gas phase products. *51st Lunar and Planetary Science Conference*, The Woodlands, Texas, oral presentation, Mar. 2020.

Wieman, S.T., J.C. Stern, S. Toler, R.T. Short. *In situ* measurements of δ^{13}C and δ^{18}O of dissolved CO_2 using an underwater mass spectrometer. *Harsh Environments Mass Spectrometry Workshop*, Myrtle Beach, South Carolina, poster presentation, Sep. 2019.

Wieman, S.T., S. Andrejkovičová, J.C. Stern. Using Isotopic Changes to Measure Modifications of Clays for Martian Habitability Studies. *Advances in Stable Isotope Techniques and Applications Workshop*, Winnipeg, Manitoba, oral presentation, Jun. 2019.

Wieman, S.T., J. Racela, D.P. Dethier. Canopy-surface cycling of chloride in a northeastern hardwood forest. *Geological Society of America Northeastern Section Meeting*, Portland, Maine, oral presentation, Mar. 2019.

Wieman, S.T., M.S. Ugelow, M.G. Trainer, J.C. Stern, M.C. Roach, J.A. Sebree. Aerosol Formation as a Fractionation Pathway in Titan's Atmosphere. *Advances in Stable Isotope Techniques and Applications Workshop*, Washington, District of Columbia, oral presentation, Jun. 2018.

Wieman, S.T., J. Racela, D.P. Dethier. Thirty-Year Trends in Acid Deposition and Mass Balance in Two Headwater Catchments, NW Massachusetts. *Geological Society of America Northeastern Section Meeting*, Bretton Woods, New Hampshire, oral presentation, Mar. 2015.

Teaching Experience

MIT-WHOI Teaching Development Fellow, Massachusetts Institute of Technology, Cambridge, MA 2024-2025

Courses Taught

Massachusetts Institute of Technology, Cambridge, MA and Woods Hole Oceanographic Institution, Woods Hole, MA

- 12.S597: Special Seminar on Teaching in Oceanography, *Course Designer, Instructor of record* 2025
- 12.747: Modeling, Data Analysis, and Numerical Techniques for Geochemistry, *Teaching Assistant* 2025
- 12.387: People and the Planet: Environmental Governance and Science, *Teaching Assistant* 2022
- WHOI Summer Math Review: Ordinary Differential Equations I, *Instructor* 2023, 2024, 2025

Harvard University, Cambridge, MA

- SPU 25: Energy and Climate: Vision for the Future, *Teaching Fellow* 2017
- PHYSCI 11: Foundations and Frontiers of Modern Chemistry, *Teaching Fellow* 2017
- SPU 29: The Climate-Energy Challenge, *Teaching Fellow* 2016

Williams College, Williamstown, MA

- GEOS 214: Geographic Information Systems and Remote Sensing, *Teaching Assistant* 2014
- CHEM 255: Intermediate Organic Chemistry Special Section, *Teaching Assistant* 2012

Teaching Coursework Taken

Graduate Teaching Certificate, Massachusetts Institute of Technology, Cambridge, MA

- Inclusive Teaching Track 2024
- Subject Design Track 2023
- Microteaching Track 2022
- Lesson Planning Track 2022

Massachusetts Institute of Technology, Cambridge, MA and Woods Hole Oceanographic Institution, Woods Hole, MA

- 12.910: Communicating Ocean Science 2024
- 12.S597: Teaching in Earth Science 2022

Professional Activities, Service, and Outreach

International

Oral Session Moderator, Ocean Sciences Meeting, Glasgow, Scotland 2026
Poster Session Moderator, Ocean Sciences Meeting, Glasgow, Scotland 2026
Managing Editor, Journal of Emerging Investigators 2021-2022
Reviewer, Journal of Emerging Investigators 2015-2022
Senior Editor, Journal of Emerging Investigators 2019-2021
TechGirls Legacy International NASA Visit, Greenbelt, MD 2019
Associate Editor, Journal of Emerging Investigators 2015-2019

National

Invited Juror, United States Invitational Young Physicists Tournament, various locations 2019-2022, 2025
Field Trip Science Lead, Death Valley Dark Skies Festival, Stovepipe Wells, CA 2020
Sample Analysis at Mars Presenter, Death Valley Dark Skies Festival, Stovepipe Wells, CA 2020
Roy J. Shlemon Mentor in Applied Geoscience, Geological Society of America, Portland, ME 2019
John Mann Mentor in Applied Hydrogeology, Geological Society of America, Portland, ME 2019
Curiosity Rover Table Presenter, Apollo 11 50th Anniversary Festival, Washington, DC 2019

Institutional

Graduate Residential Advisor, Massachusetts Institute of Technology 2023-*present*
Summer Student Fellow Mentor, Woods Hole Oceanographic Institution 2023-2025
ClimaTea Organizer, Harvard University Center for the Environment 2016
Climate Change Gallery Guide, Harvard Museum of Natural History 2017
Birds of the World Gallery Guide, Harvard Museum of Natural History 2017

Departmental

EAPS Student Advisory Committee Board Member, Massachusetts Institute of Technology 2022-2025
Geology and Geophysics Department Student Representative, MIT-WHOI Joint Program 2023-2024
Geoclub Officer, Harvard University Department of Earth and Planetary Sciences 2015-2017
Geoclub Webmaster, Harvard University Department of Earth and Planetary Sciences 2015-2016
Chemistry Student Advisory Committee Co-President, Williams College Chemistry Department 2012-2014

Local

Career Exploration Winter Event Speaker, Red Pine Scholars Program, Boston, MA 2026
Webinar Speaker, Brookline Birding Club, Brookline, MA 2025
4th Grade Erosion Activity Lead, Teaticket Elementary School, Falmouth, MA 2024
Marine Carbon Cycle Guest Lecturer, Cape Cod Community College, Hyannis, MA 2024
High School Water Chemistry Presenter, Greenwich Country Day School, Greenwich, CT 2022
Science Cafe Speaker, Science UnTapped, Woods Hole, MA 2021
5th Grade Planetary Science Presenter, Irish Elementary Escuela Bilingüe, Fort Collins, CO 2021
Seminar Series for Volunteers Speaker, Smithsonian National Air and Space Museum, Washington, DC 2020
CRESST II Undergraduate Interaction Day Presenter, NASA, Greenbelt, MD 2019
Maret School NASA Visit Host, NASA, Greenbelt, MD 2019
Chemistry Demo Day Supervisor, Williams College, Williamstown, MA 2012-2013

Professional Affiliations

American Geophysical Union 2021-*present*
Honor Society of Phi Kappa Phi 2020-*present*
Geological Society of America 2014-2022
American Chemical Society 2012-2014